The empirical recurrence for OEIS A300535, recovered from its tail

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8 September 2026

Abstract

OEIS A300535 records "Empirical recurrence of order 75" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 107$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 4$, so the recurrence is proved for every $n \geq 79$. Its last coefficient is $c_{75} = 4 \neq 0$, so no further tail satisfies anything shorter; any order-75 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture that is not written down

OEIS A300535 is "Number of nX4 0..1 arrays with every element equal to 0, 2, 3 or 4 horizontally, vertically or antidiagonally adjacent elements, with upper left element zero.". It has offset 1 and begins

$$1, 16, 36, 300, 1436, 8974, 52012, 311037, \ \ldots$$

The entry states:

> Empirical recurrence of order 75 (see link above)

As of the "Last modified" line on the live entry (Mar 08 2018 12:23:39, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \;=\; \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 107$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 107$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \ldots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S'+1$ consecutive values of a. $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 75: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 4$ is the first whose minimal order is exactly the stated 75.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 214$ exact terms from $n = 4$ on, it returns order 75 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{75} c_i\, a(n-i),$$

c_1	4	c_{20}	1882114	c_{39}	-245229168	c_{58}	-377405
c_2	22	c_{21}	6566338	c_{40}	-198097881	c_{59}	917029
c_3	-33	c_{22}	9613507	c_{41}	-56119360	c_{60}	519337
c_4	-153	c_{23}	11235378	c_{42}	67272807	c_{61}	1538222
c_5	-167	c_{24}	11110780	c_{43}	123557756	c_{62}	-672026
c_6	191	c_{25}	-1538684	c_{44}	111795070	c_{63}	99024
c_7	1323	c_{26}	-31944762	c_{45}	114244895	c_{64}	-331772
c_8	-631	c_{27}	-70963126	c_{46}	54365174	c_{65}	17734
c_9	2274	c_{28}	-95275846	c_{47}	4677084	c_{66}	16896
c_{10}	15054	c_{29}	-83821554	c_{48}	-49464797	c_{67}	31557
c_{11}	2188	c_{30}	-23460709	c_{49}	-60250951	c_{68}	15934
c_{12}	-15328	c_{31}	79762787	c_{50}	-27124944	c_{69}	-1391
c_{13}	-18663	c_{32}	216284577	c_{51}	2316553	c_{70}	-3405
c_{14}	-69296	c_{33}	285740966	c_{52}	7162586	c_{71}	-1472
c_{15}	-276997	c_{34}	214433902	c_{53}	28475058	c_{72}	40
c_{16}	-525128	c_{35}	101527520	c_{54}	4012897	c_{73}	88
c_{17}	-417815	c_{36}	-80207345	c_{55}	812845	c_{74}	12
c_{18}	-211757	c_{37}	-242879847	c_{56}	-4403316	c_{75}	4
c_{19}	-46218	c_{38}	-285319998	c_{57}	-6821165		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 79$, and it is the recurrence OEIS A300535 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 4$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{75} - c_1 x^{75-1} - \cdots - c_{75}$. Its constant term is $-c_{75} = -4$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 75 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 75, so $\deg q = 75$, and it is monic. It holds from some index t on. From $\max(t, 4)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 75, so $q = q_{\min}$. □

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 21 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 75, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 4.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-75 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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